Practical – 8

Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

A. Theory

1. Is the schema normalized properly?
Yes, the database schema is properly organized into three tables: students, events, and registrations. Each table stores specific information, reducing data duplication and maintaining data consistency.

2. Are primary keys and foreign keys correctly defined?
Yes, each table has a primary key: student_id, event_id, and registration_id, respectively. The registrations table uses student_id and event_id as foreign keys to establish relationships between students and events.

3. Is the database connection handled securely?
Yes, the database connection is handled using PHP PDO. Exception handling is used to manage connection errors, and database credentials are defined separately in db.php. This makes database connectivity easier to manage and reduces the risk of exposing connection details throughout the application.

B. Implementation/ Output/Result:

The studenthub_db database was created using MySQL in phpMyAdmin. Three tables, namely students, events, and registrations, were designed with appropriate columns and data types.

Primary keys and foreign keys were defined to maintain relationships between the tables. A unique constraint was also added to prevent duplicate event registrations.

Sample records were inserted into all three tables to test the database structure and relationships. An ER diagram was prepared to represent the entities and their connections.

A PHP file named db.php was created using PDO to establish connectivity between PHP and MySQL. Finally, a separate PHP test file was executed to verify that the database connection was working successfully.

C. Evidence of Work (Screenshots):

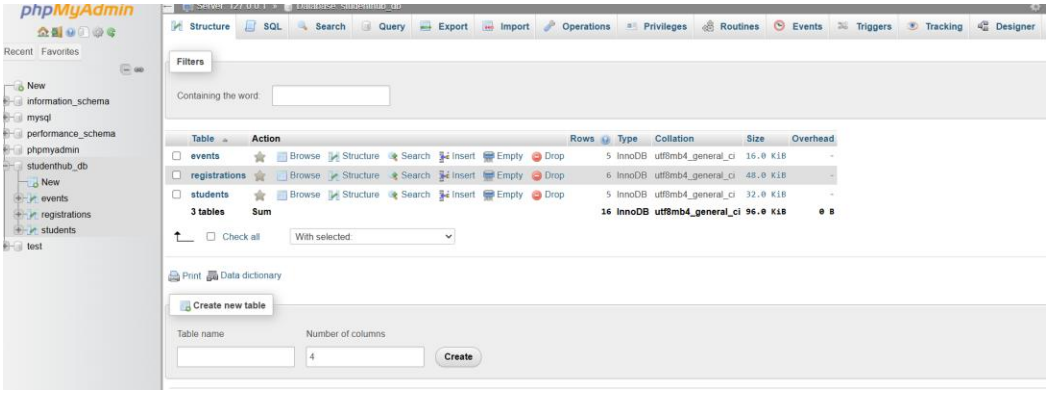


Fig1: Database in phpMyAdmin

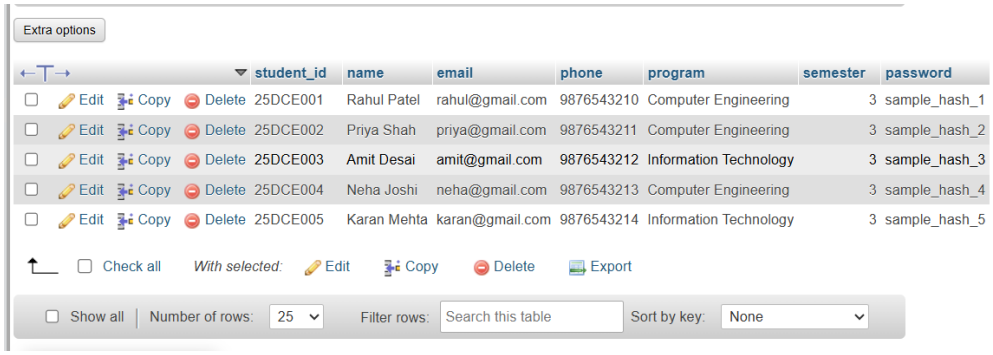


Fig2: Students table structure

Extra options

	event_id	event_name	event_date	location
Edit Copy Delete	1	Web Development Workshop	2026-11-10	CHARUSAT
Edit Copy Delete	2	Coding Competition	2026-11-15	CHARUSAT
Edit Copy Delete	3	Technical Quiz	2026-11-20	CHARUSAT
Edit Copy Delete	4	Project Exhibition	2026-11-25	CHARUSAT
Edit Copy Delete	5	Hackathon	2026-12-05	CHARUSAT

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Fig3: Events table structure

Extra options

	registration_id	student_id	event_id
Edit Copy Delete	1	25DCE001	1
Edit Copy Delete	2	25DCE001	2
Edit Copy Delete	3	25DCE002	1
Edit Copy Delete	4	25DCE003	3
Edit Copy Delete	5	25DCE004	2
Edit Copy Delete	6	25DCE005	5

Check all *With selected:* Edit Copy Delete Export

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Fig4: Registrations table structure

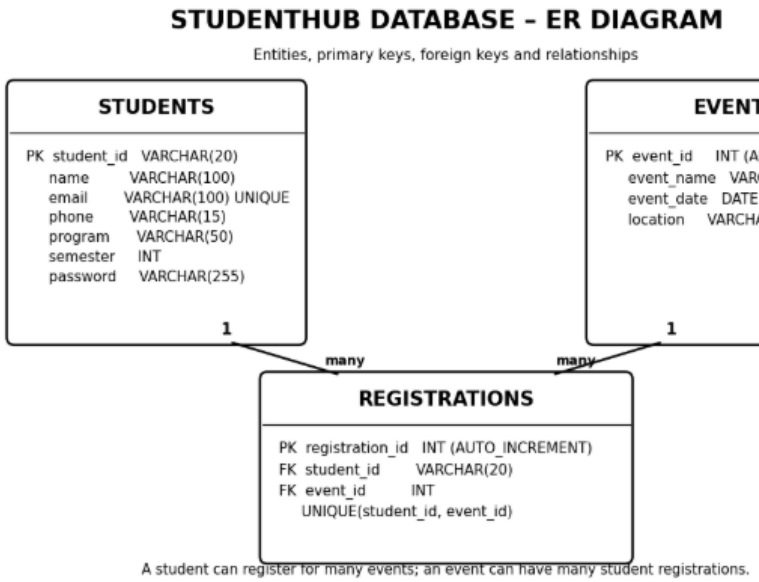


Fig5: ER diagram

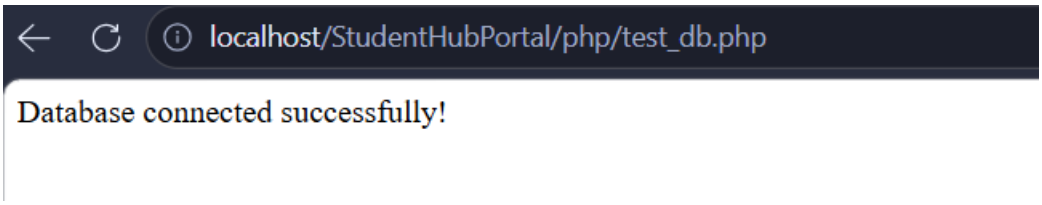


Fig6: Successful database connection

D. GitHub Repository Link:

aryan150806-wq/StudentHubPortal: Web Development Framework P

E. Challenges Faced:

1. Database Creation: Initially, setting up the MySQL database in phpMyAdmin required understanding the correct database name and configuration.

2. MySQL Port Conflict: XAMPP MySQL could not start because another MySQL service was already using port 3306. The conflict was resolved by stopping the other service.

3. Table Relationships: Defining primary keys and foreign keys between the students, events, and registrations tables required careful planning.

4. Data Insertion: Inserting sample records required ensuring that the referenced student IDs and event IDs existed before adding registrations.

5. PHP Database Connectivity: Connecting PHP with MySQL using PDO required configuring the correct database name, username, password, and connection settings in db.php

F. Learning Outcomes:

Through this practical, I learned how to create and manage a MySQL database using phpMyAdmin. I understood how to design database tables with appropriate columns, data types, primary keys, and foreign keys. I gained knowledge of database relationships and normalization through the students, events, and registrations tables. I also learned to insert and retrieve records using SQL queries and represent table relationships using an ER diagram. Additionally, I gained practical experience in connecting PHP with MySQL using PDO and testing the database connection. This practical helped me understand how databases are used to store and manage information in a web application.

Faculty Remarks: